

BATNA, Algeria

WMO: 604680

Lat: 35.752N

Lon: 6.309E

Elev: 2697

StdP: 13.32

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.5	28.6	19.5	16.2	47.3	22.1	18.3	42.8	32.2	50.4	28.1	49.3	2.9	120	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.5	102.5	65.9	100.1	65.3	97.0	65.0	70.2	90.5	68.6	88.8	67.4	87.6	16.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	100.7	76.1	62.8	94.5	74.8	61.1	88.9	73.6	35.9	90.9	34.4	88.5	33.3	87.5	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.7	24.0	21.1	DB	21.8	106.9	2.3	1.4	20.2	108.0	18.9	108.8	17.6	109.6	16.0	110.7
			WB	21.0	73.4	2.6	2.9	19.1	75.5	17.5	77.2	16.0	78.8	14.1	81.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	43.2	44.5	50.5	56.3	64.9	74.2	81.0	79.4	71.1	63.0	51.7	45.0	(6)
(7)		DBStd	14.48	4.58	5.60	6.31	6.44	7.27	7.00	5.25	5.01	6.06	6.18	5.76	4.93	(7)
(8)		HDD50	689	216	171	71	15	0	0	0	0	1	47	168		(8)
(9)		HDD65	3216	675	575	449	270	93	9	0	14	111	401	619		(9)
(10)		CDD50	4524	6	15	88	204	463	726	960	913	634	405	96	14	(10)
(11)		CDD65	1576	0	0	1	8	90	285	495	448	198	50	1	0	(11)
(12)		CDH74	21632	0	5	48	240	1433	4012	7023	5984	2208	659	19	1	(12)
(13)		CDH80	11875	0	1	8	45	575	2182	4343	3543	973	204	1	0	(13)
(14)	Wind	WSAvg	8.2	7.4	8.5	8.9	8.8	8.6	8.7	9.0	8.4	7.8	7.4	7.7	6.9	(14)
(15)	Precipitation	PrecAvg	14.4	1.6	1.2	1.5	1.6	1.7	0.6	0.3	0.8	1.4	1.1	1.3	1.5	(15)
(16)		PrecMax	23.9	5.5	4.0	3.3	4.0	4.1	2.0	1.3	2.9	3.0	3.5	4.1	5.0	(16)
(17)		PrecMin	8.9	0.3	0.0	0.2	0.2	0.1	0.0	0.0	0.1	0.4	0.3	0.4	0.0	(17)
(18)		PrecStd	3.6	1.2	0.9	0.7	0.9	1.2	0.5	0.3	0.8	0.7	0.8	0.9	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	71.6	79.1	84.5	96.0	102.7	106.1	104.8	96.5	89.8	76.8	67.9	(19)
(20)		0.4%	MCWB	49.6	50.5	54.3	57.5	61.1	64.2	67.5	65.8	65.5	61.4	56.9	52.2	(20)
(21)		2%	DB	61.2	65.8	73.6	80.6	90.0	99.0	103.8	101.9	93.1	85.6	71.4	62.6	(21)
(22)		2%	MCWB	48.4	49.5	52.4	56.3	60.9	63.7	67.0	65.6	64.6	60.1	53.9	50.1	(22)
(23)		5%	DB	57.4	61.0	69.8	76.4	86.4	95.9	100.7	98.8	89.6	81.0	67.8	59.0	(23)
(24)		5%	MCWB	46.7	47.2	51.6	55.4	60.1	63.3	65.9	65.1	63.9	59.4	52.6	48.6	(24)
(25)		10%	DB	53.9	57.2	65.8	71.8	82.5	91.8	98.2	95.8	86.0	76.9	64.2	55.8	(25)
(26)		10%	MCWB	45.2	45.8	50.2	54.2	59.3	62.9	65.5	65.2	63.9	58.7	51.6	47.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	52.9	56.6	61.8	66.5	68.7	73.3	72.6	70.8	66.0	59.5	53.9	(27)
(28)		0.4%	MCDWB	62.5	64.7	71.5	77.9	83.5	91.1	95.0	92.3	85.7	80.4	69.7	64.9	(28)
(29)		2%	WB	49.4	50.4	54.4	59.0	64.2	66.7	70.3	70.1	68.3	63.4	56.6	51.2	(29)
(30)		2%	MCDWB	59.0	62.7	68.9	72.4	81.5	88.6	93.2	88.8	83.0	76.9	66.4	59.8	(30)
(31)		5%	WB	47.5	48.3	52.7	57.1	62.2	65.4	68.7	68.4	66.6	61.7	54.3	49.4	(31)
(32)		5%	MCDWB	56.2	59.1	66.5	71.4	79.4	87.2	91.7	87.8	81.8	74.7	64.6	57.3	(32)
(33)		10%	WB	45.8	46.6	51.0	55.4	60.6	64.1	67.3	67.2	65.3	60.2	52.5	47.7	(33)
(34)		10%	MCDWB	53.2	55.7	63.7	68.9	77.4	85.5	90.5	86.9	80.3	72.5	62.0	54.7	(34)
(35)	Mean Daily Temperature Range		MDBR	20.9	22.4	24.5	25.0	27.3	29.3	31.5	30.3	25.8	24.5	22.0	20.4	(35)
(36)		5% DB	MCDBR	28.3	30.8	32.3	32.9	33.0	33.2	34.1	33.9	30.3	30.6	28.1	26.9	(36)
(37)			MCWBR	17.4	17.2	15.9	14.4	12.6	10.5	11.1	11.2	10.8	12.3	15.1	16.6	(37)
(38)		5% WB	MCDWB	25.9	27.5	29.4	28.5	30.2	31.3	31.2	30.3	26.6	25.3	24.9	23.9	(38)
(39)			MCWBR	16.7	16.2	15.6	14.0	12.9	10.9	11.5	11.4	10.6	11.6	14.3	15.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.289	0.311	0.371	0.406	0.446	0.442	0.486	0.460	0.430	0.381	0.330	0.298	(40)	
(41)		taud	2.511	2.447	2.261	2.181	2.087	2.121	1.987	2.063	2.161	2.289	2.434	2.505	(41)	
(42)		Ebn at Noon	292	296	284	278	268	268	255	259	261	265	272	280	(42)	
(43)		Edh at Noon	26	31	40	46	51	50	56	51	44	36	28	24	(43)	
(44)	All-Sky Solar Radiation	RadAvg	837	1099	1483	1821	2101	2356	2383	2085	1635	1263	911	749	(44)	
(45)		RadStd	85	90	111	135	112	87	60	87	72	71	56	71	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	+0.70	+0.70	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page